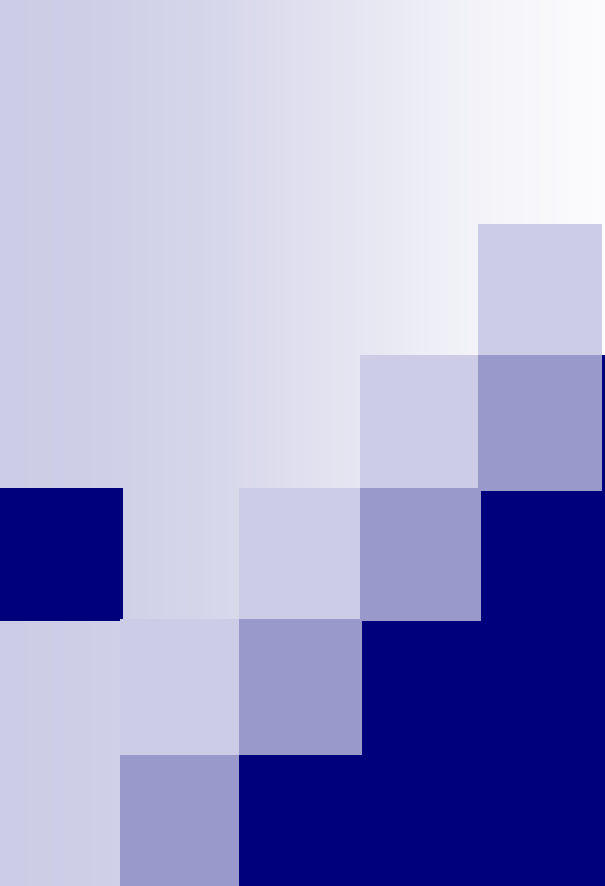




# Software Quality Assurance (CS-706)



## Lecture # 1

Dr. Ghulam Ahmad Farrukh

# Introduction - 1

- This course deals with a very important aspect of software engineering: quality assurance of software products and services
- We'll learn different aspects of software quality assurance in this course

# Introduction - 2

- In the first few lectures, we will discuss what software quality is and how it impacts the software development and maintenance and other basic concepts in SQA
- In the second phase of this course, we'll discuss in detail the activities in each phase of the software development lifecycle, as they relate to software quality assurance

# Introduction - 3

- In the third part of this course, we'll discuss different topics related to software quality assurance. We'll look at quality assurance processes, some of the major process improvement programs from the quality assurance' perspective
- We'll also study some other topics, given our time constraints

# What is Quality?

- Can you define quality?
- You must be thinking, what kind of question is that. It is very easy to define quality, but if you think it is really hard it is not that easy to define quality
- Have you come with a definition? Let's see what I have in store for you

# Synonyms of Quality

- Excellence
- Superiority
- Class
- Eminence
- Value
- Worth



# Antonym of Quality

- Inferiority



# Marketability of Quality

- Everyone claims to manufacture / develop / sell / market “good” quality products / services
- You will never come across a person or company selling products or services as low-quality or poor-quality products, even when they are

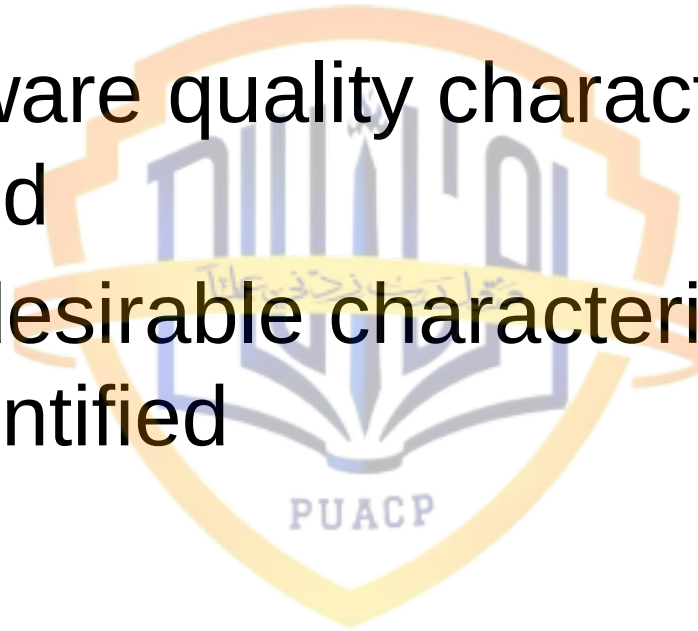


# Software Quality - 1

- Quality as it relates to all aspects of software (requirements / design / code / tests / documents / training)
- Difficult to define
  - Software quality is somewhat like the concept of beauty. Each of us has a strong opinion about what constitutes beauty, and we recognize it when we see it. But when asked to explain exactly why we regard an object as beautiful, it is hard to put the factors into words

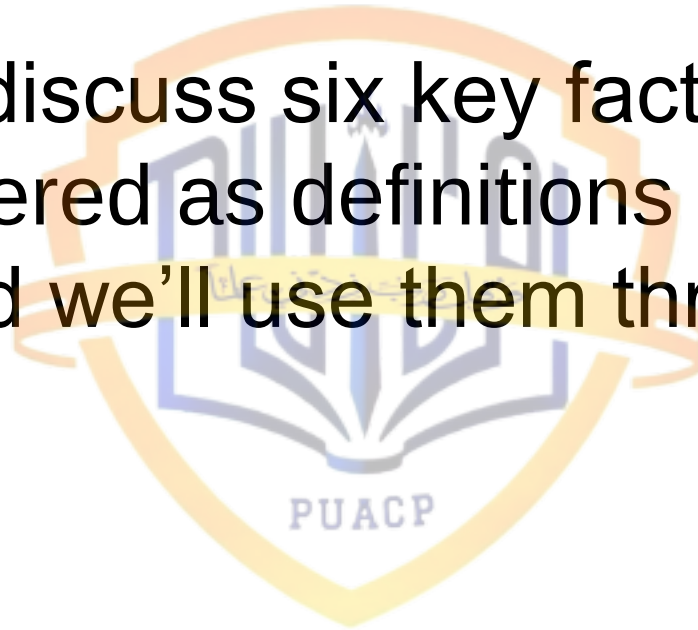
# Software Quality - 2

- Good software quality characteristics can be identified
- Bad or undesirable characteristics can also be identified



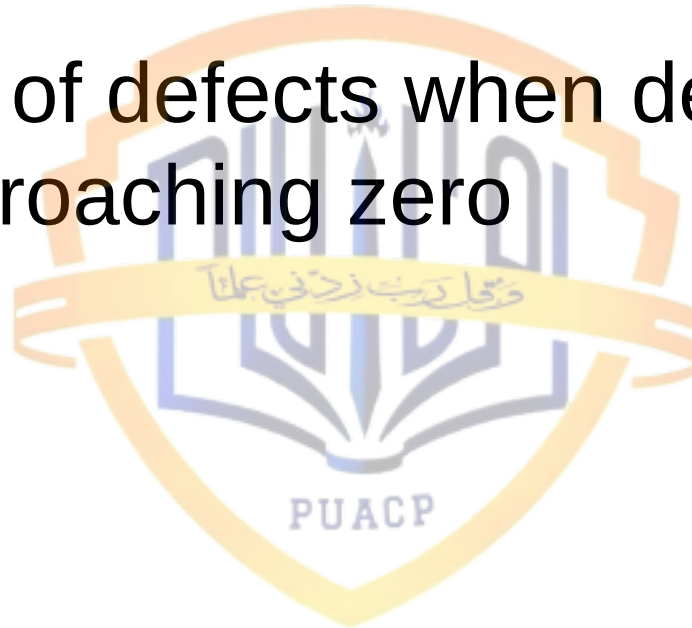
# Software Quality Definitions

- Now we'll discuss six key factors, which are considered as definitions of software quality, and we'll use them throughout this course



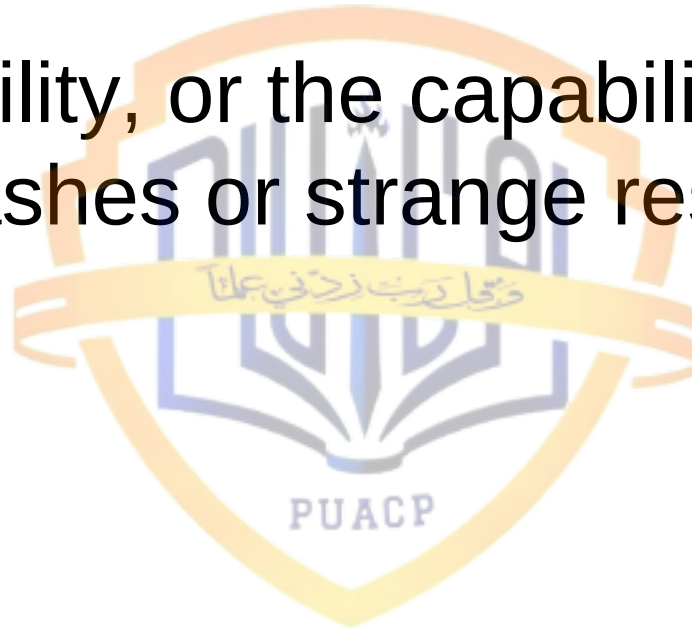
# Software Quality

- Low levels of defects when deployed, ideally approaching zero



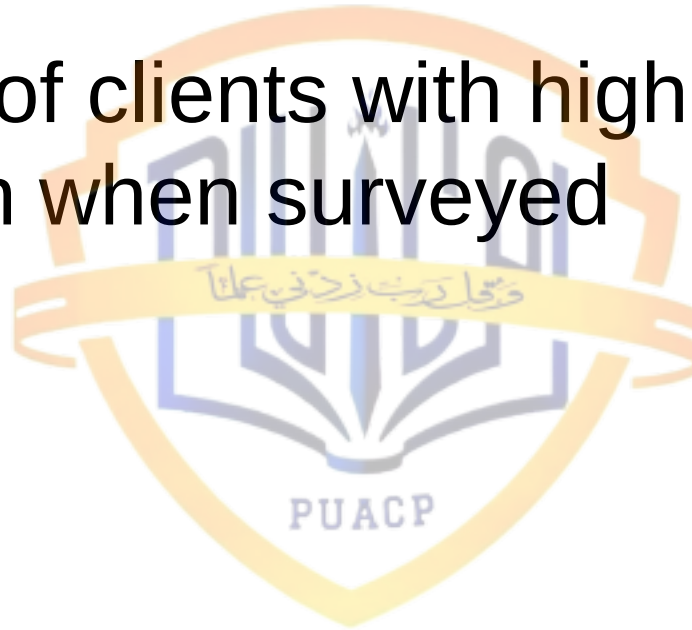
# Software Quality

- High reliability, or the capability of running without crashes or strange results



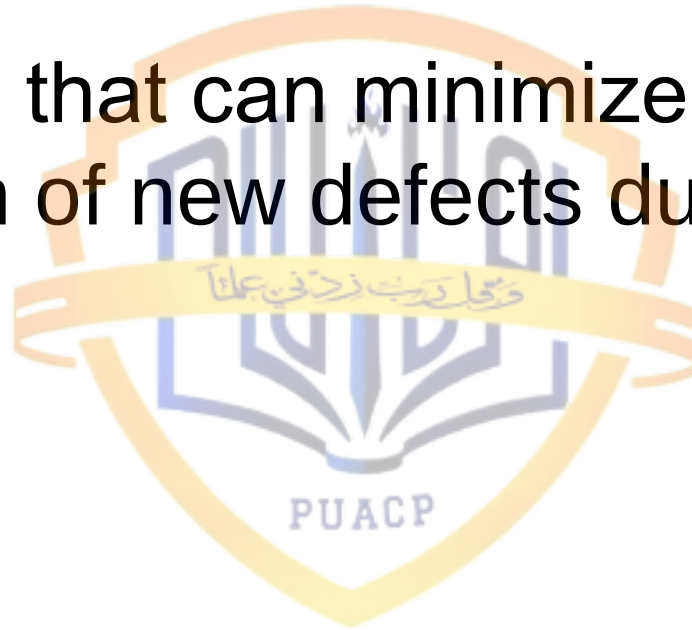
# Software Quality

- A majority of clients with high user-satisfaction when surveyed



# Software Quality

- A structure that can minimize “bad fixes” or insertion of new defects during repairs



# Software Quality

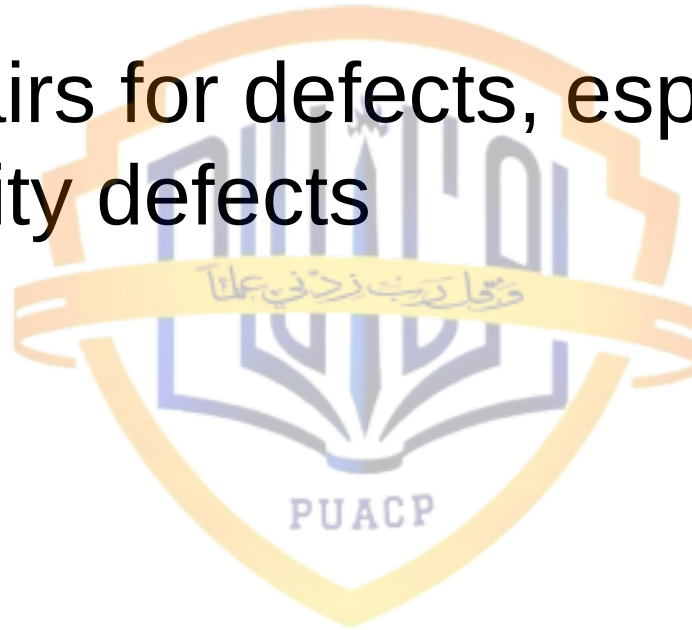
- Effective customer support when problems do occur





# Software Quality

- Rapid repairs for defects, especially for high-severity defects



# Why Software Quality? - 1

- Reduces time to market for new products
- Enhances market share compared to direct competitors
- Minimizes “scrap and rework” expenses
- Attracts and keeps “top-gun” personnel
- Minimizes the risk of serious litigation

# Why Software Quality? - 2

- Minimizes the risk of serious operating failures and delays
- Minimizes the risk of bankruptcy or business failures, which may be attributed directly to poor quality or poor software quality

# Software Quality Assurance

- So the term software quality assurance would mean that the software guarantees high quality
- In this course, we'll learn the different processes, techniques, and activities, which enables us – the software professionals – to provide that guarantee to ourselves and our clients

# Achieving Software Quality

- “For a software application to achieve high quality levels, it is necessary to begin upstream and ensure that intermediate deliverables and work products are also of high quality levels. This means that the entire process of software development must itself be focused on quality”
  - Capers Jones

# Summary

- In today's lecture, we have only discussed what quality is and what software quality is
- We have briefly touched upon the need of software quality
- In the coming lectures, we will explore software quality assurance in quite a bit of detail, so get ready for a very exciting course

# References

- Software Quality: Analysis and Guidelines for Success by Capers Jones

